Module 07 Lab Exercise - Contribution Journal

John Grubbs

I completed and ran my own copy of the Module 7 Model Evaluation lab using the Titanic dataset. I changed the Sex column into numbers, filled in missing Age values with the median, and used Age, Pclass, Sex, and Fare to predict survival with Logistic Regression.

I created a confusion matrix and classification report to review the model's predictions. The model reached about 80% accuracy. I also used five-fold cross-validation. The average score was 78.68%, with a standard deviation of 0.0094. I completed the knowledge check by explaining precision, recall, and why cross-validation is useful.

Keira Burdine

During Module 7, I worked through the model evaluation notebook in Google Colab. I reviewed the provided setup code, corrected an issue with missing Age values and successfully trained a logistic regression model using the Titanic dataset. I ran each code cell and checked the results before continuing.

I generated and visualized a confusion matrix using scikit-learn and Seaborn. I then created a classification report to examine precision, recall, F1 score, support and overall accuracy. Finally, I used cross_val_score to perform 5-fold cross-validation and printed the fold scores, average score and standard deviation.

I interpreted the model results rather than only producing the outputs. I identified 144 correct predictions out of 179 test cases and compared the model's performance for survivors and non-survivors. I also reviewed real world examples of when precision or recall matters more and explained why cross-validation is generally more dependable than relying on a single train-test split.

My contribution was completing the notebook tasks, troubleshooting errors, verifying the generated results and preparing written explanations of what the evaluation metrics mean. This lab strengthened my practical skills in Python, Google Colab, model evaluation and interpreting machine learning performance.

Luis Adrian

For this lab, I worked on completing the Python notebook using Google Colab. I used the Titanic dataset and trained a Logistic Regression model to predict passenger survival. I completed the code for generating a confusion matrix and used Seaborn to visualize the results. I also generated a classification report to evaluate the model using Precision, Recall, and F1-Score. These tasks helped me to understand how to identify correct and incorrect predictions.

I also worked on the Cross-Validation section, where I used 5-fold cross-validation to evaluate the model multiple times. I calculated the average accuracy of 78.68% and the standard deviation of 0.0094. After completing the coding sections, I answered the Knowledge Check questions using examples to explain the differences between Precision and Recall and why Cross-Validation is important.

My contribution to this lab focused on completing the assigned coding tasks, reviewing the model's results, and explaining what I learned from the evaluation process. This experience helped me improve my Python skills and my understanding of Machine Learning evaluation techniques.